C11 Introduction to Organic Chemistry – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Key Definitions **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Hydrocarbon				Compound of hydrogen and carbon only
2	Saturated				Only single C–C bonds
3	Unsaturated				Contains a C=C double bond
4	Homologous series				Same functional group and general formula; each member differs by CH_2
5	Functional group				Group of atoms responsible for a compound's characteristic reactions
6	General formula				Algebraic formula for any member of a homologous series
7	Molecular formula				Actual number of atoms of each element
8	Structural formula				Shows arrangement of atoms, e.g. $CH_3CH_2CH_3$
9	Displayed formula				Shows every atom and every bond
10	Skeletal formula				Carbon skeleton and functional groups only; H on C not shown

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C11 Introduction to Organic Chemistry – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 2: Isomerism **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Structural isomers				Same molecular formula, different structural formula
2	Three types of structural isomerism				Chain, position, functional group
3	Stereoisomers				Same structural formula, different spatial arrangement
4	Why E–Z isomerism occurs				Restricted rotation about C=C
5	Requirement for E–Z isomerism				Two different groups on each C of the C=C
6	*Z* isomer				Higher-priority groups on the same side
7	*E* isomer				Higher-priority groups on opposite sides
8	CIP priority is based on				Atomic number
9	Functional group isomer of propanal				Propanone
10	Number of structural isomers of C_4H_{10}				2

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 3: Naming: Stems and Suffixes **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Stem for 1 carbon				meth-
2	Stem for 2 carbons				eth-
3	Stem for 3 carbons				prop-
4	Stem for 4 carbons				but-
5	Stem for 5 carbons				pent-
6	Stem for 6 carbons				hex-
7	Suffix for an alkane				-ane
8	Suffix for an alkene				-ene
9	Suffix for an alcohol				-ol
10	Suffix for an aldehyde				-al

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 4: Naming: Functional Groups and Rules **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Suffix for a ketone				-one
2	Suffix for a carboxylic acid				-oic acid
3	Prefix for a chlorine substituent				chloro-
4	Prefix for a bromine substituent				bromo-
5	Name of the $-CH_3$ substituent				methyl
6	Prefix for two identical substituents				di-
7	Prefix for three identical substituents				tri-
8	Order of substituent prefixes				Alphabetical
9	Rule for numbering the carbon chain				Lowest possible numbers
10	Prefix for a ring of carbons				cyclo-

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C11 Introduction to Organic Chemistry – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 5: Functional Groups and General Formulae **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Alcohol functional group				–OH
2	Aldehyde functional group				–CHO
3	Ketone functional group				C=O within the chain
4	Carboxylic acid functional group				–COOH
5	Alkene functional group				C=C
6	Halogenoalkane functional group				C–X
7	General formula of alkanes				C_nH_{2n+2}
8	General formula of alkenes				C_nH_{2n}
9	General formula of cycloalkanes				C_nH_{2n}
10	General formula of alcohols				$C_nH_{2n+1}OH$

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.